

Exploring the Trade-offs between Energy and Performance of Federated Learning Algorithms

— PhD Progress Report 2nd year —

Mai Huong Do

SEPIA team
IRIT - Université Toulouse

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Supervisors: Georges Da Costa, Millian Poquet

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Thesis Introduction

Energy Consumption in Federated Learning (FL)

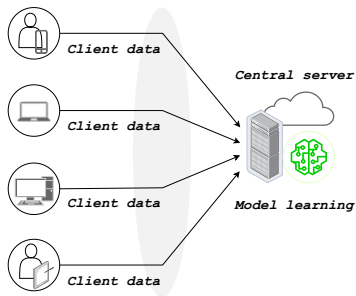


Fig 1. Classic Machine Learning: Data Comes to the Model → Data privacy issues.

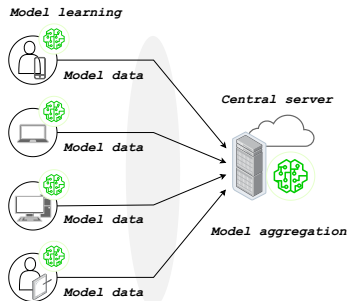


Fig 2. Federated Learning (FL)
Model Goes to the Data.

- **Computing energy cost:** From centralized computing to distributed client-side computation.
- **Communication energy cost:** From one-time heavy data transfer to frequent lightweight model updates.

- This thesis is one part of **ANR DELIGHT** (aDvancing fEderated LearnIng while reducinG tHe carbon fooTprint) project.
- Target of **DELIGHT**: improve FL performance but also incorporate **energy efficiency** as **one of the metrics of FL** to push FL towards sustainability.
- Target of **my thesis**: develop an automatic framework to explore the trade-offs between Energy and FL performance.

Phases of the thesis:

- Set up an experimental environment on Grid'5000 ¹ (g5k) to gather performance and energy metrics.
 - Create a use-case for the Flower framework.
 - Build a reproducible and automated framework for obtaining metrics for this use case.
- Propose, formulate energy model, and implement the different leverages. **(current)**
- Explore the impact of the leverages on both energy and performance. **(current)**

¹a distributed testbed system (multiple clusters, nodes, cores,...).

Thesis Progress

Last year - 1st

Last Year's Committee Meeting - 1st year

From the beginning to the end of August 2024, I:

- Conducted a survey of related work from 2019 to 2024 (31 papers).
- Carried out initial experiments on energy measurement in a distributed system (Grid'5000).

The future work outlined last year (for 2nd year) included:

- Keep updating the survey.
- Make the measurement in automating & improving stability.
- Test with other FL strategies, multiple test case.
- Propose a framework to monitor energy consumption & FL performance.

This year

This year - 2nd

What did I do?

- A **survey** with 50 articles (selected from 263 screened articles), focus on energy and experimental design for FL.
- **FedEator**: framework measuring the energy consumption of FL in distributed systems.

- **Measuring Communication Energy:** Diverse protocols and hardware complicate accurate communication energy assessment.
- **Neglect of Carbon Footprint:** Environmental impact is often overlooked; carbon-aware FL models are needed.
- **Energy–Performance Tradeoff:** Balancing energy savings and model accuracy remains a fundamental challenge.
- **Lack of Real-World Experiments:** Most studies rely on simulation; real testbed validation remains rare but essential.
- **Lack of standardization:** No standard energy measurement framework for FL; results are hard to compare and validate.

► Articles Scanning - see more

Current energy measurement methods (in general):

- **Hardware-based:** External wattmeters measure total device energy.
 - Accurate but coarse-grained and inflexible.
- **Software-based:** Tools like Mojito/S, Perfetto, PowerAPI, Alumet ² estimate per-component energy via internal sensors.
 - More detailed, but often less accurate under heavy load.

²found from google search engine and articles

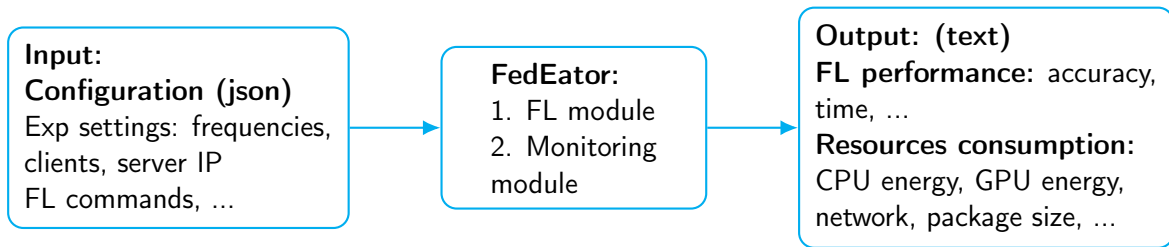
Propose FedEator:

- A framework for system resources monitoring (energy, power, network,...), synchronized software + hardware based methods.
- Supports DVFS (dynamic voltage frequency scaling).
- One command → multiple settings × repeated runs → reproducibility & stability check.
- Validated on Grid'5000 system.



Framework Description

Framework Description

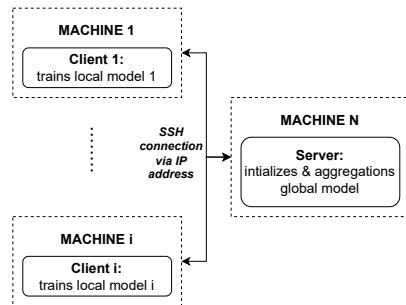


Framework Description

1. FL module

Manage the FL processes:

- Clients running
- Server running
- FL performance logging



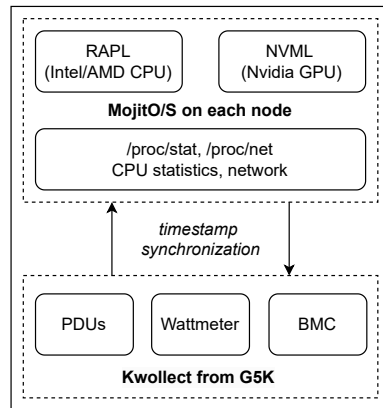
Framework Description

2. Monitoring module

System state and energy consumption records, sync between software^a and hardware^b.

^ain Joules, sampling rate from 10 Hz to 100 Hz

^bin Watts, sampling rate from 0.2 Hz to 1 Hz



Framework Validation

Testbed: Grid'5000 (taurus, gros, chiffloth clusters)

Scenarios:

- **CPU nodes:** taurus-1,8,9,13 – DVFS 4 frequencies (min, max, 2 in between)
- **GPU nodes:** chiffloth-3,4,5,6 – DVFS maximum frequency
- **Large-scale:** 20 machines from gros (1 server, 19 clients) - DVFS 4 frequencies
- CPU test cases run twice while GPU and large scale run once.

FL Configuration:

- **ML task:** Image classification
- **Model:** MobileNetV2, batch size=32
- **Dataset:** CIFAR10, distributed across clients, train/test = 80/20
- **Training:** Local epochs {1,2}, 15 rounds, FedAvg and FedAvg-customized aggregation

Experiments Results: CPU Nodes

- RAPL (CPU) vs wattmeter → consistent trends.
- Energy spikes correspond to FL rounds.
- The power meter readings are slightly higher due to measuring full-system power, beyond what RAPL covers.

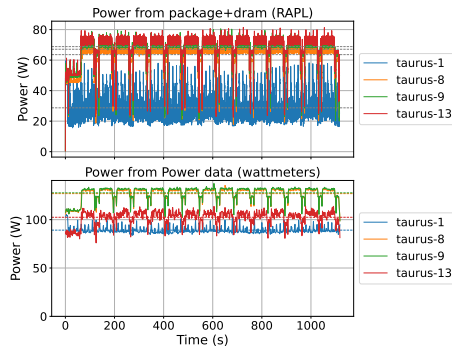


Fig. 3. Power consumption measured by RAPL and an external power meter; data are converted to Watts, 1/10 second intervals. (FedAvg, 2 epochs, $f=1.6\text{GHz}$)

Experiments Results: CPU Nodes

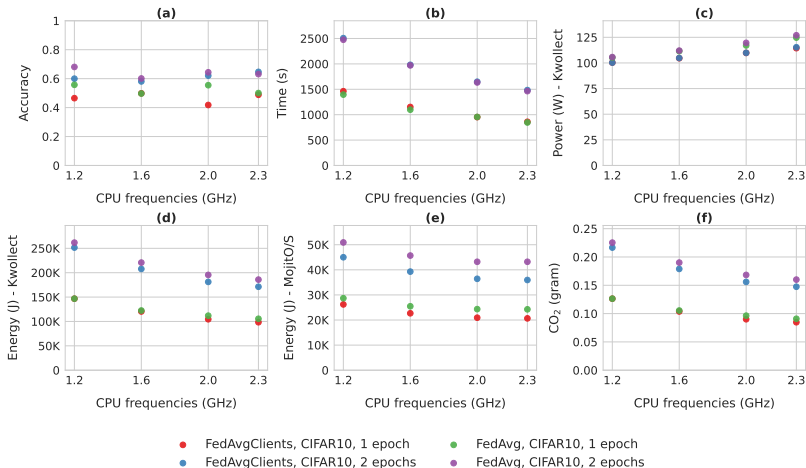


Fig 4. FL performance and resource consumption under 4 CPU frequency settings for 3 CPU clients + 1 server; each point represents the mean of two repetitions. (f)^a

^aCO₂ emission estimation based on French carbon intensity (31 gCO₂eq/kWh)

Experiments Results: Large-scale

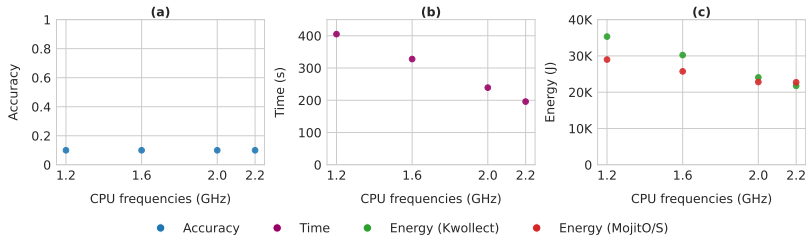


Fig 5. FL performance and resource consumption under 4 CPU frequency settings, shown for a single FL instance configuration (FedAvg, CIFAR-10, 1 local epoch), for 19 CPU clients.

Experiments Results: GPU Nodes

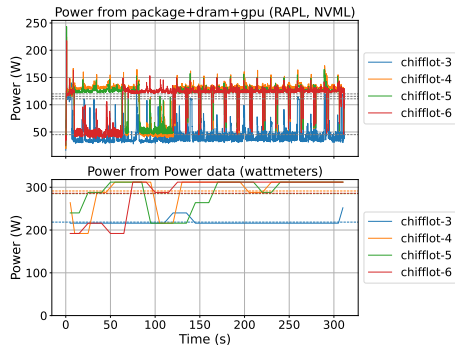


Fig 6. Power consumption measured by RAPL + NVML and an external power meter; data are converted to Watts, 1 second intervals. (FedAvg, 2 epochs, $f=1.328\text{GHz}$)

TABLE II
RESULTS OF CONFIGURATIONS: FedAvg, CIFAR-10, GPU FREQUENCY 1328 MHz, IN TWO SETTING OF ONE OR TWO EPOCH OF LOCAL TRAINING PER ROUND IN **GPU NODES** EXPERIMENT.

Number of epoch	1	2
Total energy by RAPL+NVML (J)	14525	20165
Total energy by wattmeters (J)	58643	80702
Accuracy	0.4903	0.6692
Total time (s)	212.16	298.58
CO ₂ (gram)	0.050	0.069

Experiments Results: Discussion

- Framework successfully captures distributed energy patterns.
- Relationship between CPU/GPU frequency and energy consumption:
Higher frequencies $\rightarrow$ less time processing but more power $\rightarrow E = P \times t$?
- Difference between CPU and GPU:

Local epochs	1 epoch		2 epochs	
Processor	CPU	GPU	CPU	GPU
Freq_max (GHz)	2.3	1.328	2.3	1.328
Energy RAPL (J)	25K	14K	43K	20K
Energy Wattmeter (J)	100K	58K	180K	80K
Accuracy	0.49	0.49	0.67	0.67
Time	900	212	1500	298
Co2	0.09	0.05	0.16	0.07

Next year - 3rd

My plan for the upcoming year includes:

- **Reproducibility:** Package the environment with Nix for reliable experiment replication.
- **Experiment extension:** Test more FL strategies, datasets, and tasks to assess accuracy–energy trade-offs.
- **Generality:** Support custom node connections across distributed systems, not just Grid'5000.
- **Optimization:** Integrate resource allocation and optimization algorithms with energy measurement (if possible).
- **And finally:** Submit publications and start writing the thesis.

Work Situation

Difficulties Encountered

- No major difficulties; work is progressing as planned.
- Need to quickly revise and submit two manuscripts.
- The biggest challenge: the harsh sunlight!

- The **FedEator** framework, which is available as open-source here:
<https://gitlab.irit.fr/sepia-pub/delight/eflwr>
- Two manuscripts under revision for resubmission:
 - A **survey paper**
 - A paper describing the **FedEator framework**
- Presented my work and had a paper accepted at the **COMPAS 2025** conference.



Collab with DELIGHT Project Members

Name	Team	Main Task (Phase 1)
Ahmad	LIA	Implement optimal client selection (OptFed) in Flower
Oumaima	LAAS	Implement and assess optimal sampling and data summarization in OptFed scenarios
Mai	IRIT	Energy consumption experiments with 100+ clients; run all scenarios after Ahmad and Oumaima complete their parts

Tutorials for TP and TD sessions of the CSA class in the *Parallel* course (14h):

- **Monitoring**
- **MPI**

This year, I continue to take on these classes.

Completed Training Activities:

- *Débat théâtral "Entre les lignes"* (April 2025) — 3 hours
- *Journée Doctorants ASR* (June 2024) — 8 hours
- *MOOC: Reproducible Research II* (June-September 2024) — 10 hours
- *MOOC: Recherche reproductible* (September 2023-July 2024) — 6 hours
- *Cours de Français Langue Étrangère – FLE* (September 2024-June 2025) — 23 hours (Grade: 14)
- *EcolCT Fall School 2024* (October 2024) — 20 hours
- *What do we mean by Digital Sobriety?* (October 2024) — 2 hours
- *PhDOOC – MOOC Doctorat et Poursuite de Carrière* (October 2024-March 2025) — 12 hours

Total training hours: 84/100 hours

Merci pour votre attention!

Questions et discussion

Appendix: Articles Scanning Flowchart

